CSA0711 - Computer Networks

ASSIGNMENT 1

CO1: Apply the functions of the OSI layers in enabling reliable data transmission across networks – uses of networks, network hardware and software, components, types of connections, topologies, protocols and standards, the ISO/OSI reference model,

transmission media, wireless communication, Ethernet configuration, an introduction to 5G networks, and error detection, error correction and data-link control.

Assignment Title: Make the Data of a Smart Manufacturing Plant Arrive Correctly and On Time (Total Marks - 50)

Nature of Assignment: Individual

Duration: 1 Week

Submission Date: 12/11/2026

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Course Code: CSA0711

Course Name: Computer Networks

Bloom's Levels Assessed: K5 – Evaluate (every task asks you to calculate or analyse first, and then judge, decide or justify)

Mandatory Execution and GitHub Submission

• Every student must individually execute every task in this assignment, including all "Complete the Code" programs.

• Create a GitHub repository named CSA07-<RegisterNumber>Assignment1 and share it with the course faculty (public, or private with the faculty added as a collaborator).

• Keep one folder for each task (for example: Task01, Task02 … or Q1, Q2 …), containing the Python programs, Packet Tracer (.pkt) files, device configurations, Wireshark captures, command outputs and screenshots.

• Include a README.md with your name, register number, the operating system, Python version and Packet Tracer (or NS-3) version used, the steps to run each program or open each simulation, and a short note of the result of each task.

• Commit your work progressively – at least one commit for each task. The commit history will be checked; a single upload of all files on the last day will not be accepted.

• Submit the GitHub link below and on the cover page of your report. An assignment submitted without a working GitHub link will not be evaluated.

GitHub Repository Link: [Computer-Networks-Lab/Assignment 1 at main · KarthikMG-gf/Computer-Networks-Lab]

How to Answer the Additional Questions

Complete the Code: Copy the given program or device configuration into your repository,

replace every TODO with working code or commands without changing the rest, run it (Python 3 or Cisco Packet Tracer) and compare your output with the expected output or result.

Alternative Concept Analysis: Describe how the solution would change if the alternative

concept were used, and evaluate both solutions using network measures such as throughput, delay, jitter, packet loss, overhead, reliability, security and cost, supported by your own calculations, simulations or measurements wherever possible.

Scenario — Network for a Smart Manufacturing Plant

An automobile-parts company is building a smart plant. Robots on the shop floor must receive control commands within 10 milliseconds; hundreds of sensors report temperature and vibration; CCTV cameras stream video; office staff use the ERP system and e-mail; and the plant exchanges production data with the head office 400 km away.

The contractor proposes ordinary office Wi-Fi throughout the plant and a single broadband connection to the head office. The plant engineers are worried that electrical noise from welding machines will corrupt data, that robot commands will be delayed, and that the head-office link will be too slow.

The network may grow:

1 production line (40 devices) → 1 plant (1,500 devices) → 5 plants connected to the head office Each task below gives you data from the plant. Use it to calculate, compare and decide – a correct answer must be supported by your working, not only stated.

Design Challenge:

The company plans to use a private 5G network on the shop floor and to connect autonomous guided vehicles (AGVs). Re-evaluate your original design and determine whether it should be retained, modified or redesigned.

TASK 1 – Size the Traffic Before Choosing Anything **4 Marks**

Bloom's Level: K5 – Evaluate

One production line generates the following traffic:

Source	Quantity	Traffic
Robot controllers	40	200-byte command every 2 ms each
Vibration sensors	600	50 bytes per second each
CCTV cameras	24	4 Mbps each (continuous)
Office users	150	2 Mbps each at peak, 10% active at a time

Work out:

- The total traffic of one line in Mbps, and of the plant if there are 8 lines
- Which source sends the most data and which has the strictest delay requirement – are they the same?
- Whether a single 1 Gbps uplink from the shop floor is enough, and what happens to robot commands if the CCTV traffic shares the same queue
- A ranking of the traffic types by how harmful a 100 ms delay would be, with a reason for each

Answer

1) Total traffic.

Source	Calculation	Traffic
Robot controllers	40 × 200 B × 8 bit × 500 commands/s	32.00 Mbps
Vibration sensors	600 × 50 B × 8 bit/s	0.24 Mbps
CCTV cameras	24 × 4 Mbps	96.00 Mbps
Office users	150 × 2 Mbps × 10%	30.00 Mbps
One line total	32 + 0.24 + 96 + 30	**158.24 Mbps**
Plant (8 lines)	8 × 158.24	**1,265.92 Mbps ≈ 1.27 Gbps**

2) Largest source and strictest delay. CCTV sends the most data (96 Mbps = 60.7% of the line traffic). The robot controllers have the strictest delay requirement (a command every 2 ms and a 10 ms deadline). They are not the same source: the biggest volume is not the most delay-sensitive.

Source	Volume	Share of line	Delay tolerance
Robot controllers	32.00 Mbps	20.2%	10 ms (strictest)
Vibration sensors	0.24 Mbps	0.2%	about 1 s
CCTV cameras	96.00 Mbps	60.7%	100–500 ms (buffered)
Office users	30.00 Mbps	19.0%	100–200 ms

3) Is a 1 Gbps uplink enough? One line needs 158.24 Mbps, which is 15.8% of 1 Gbps, so it is enough. Eight lines need 1,265.92 Mbps = 126.6% of 1 Gbps, so a single 1 Gbps uplink is not enough for the plant (and even 70% of the link, 700 Mbps, would be exceeded). If CCTV shares the same queue with robot commands, the 266 Mbps of excess traffic keeps the queue full and the commands wait behind video packets: every 1 MB of queue adds 8 ms of delay and 2 MB adds 16 ms, which is already above the 10 ms deadline, and once the buffer is full, packets – including robot commands – are dropped. The robot traffic therefore needs its own VLAN and a strict-priority queue, and the uplink must be larger.

4) Ranking of the harm from a 100 ms delay.

Rank	Traffic	Reason
1	Robot commands	100 ms is 10 times the 10 ms deadline and equals 50 missed 2 ms commands; the robot moves on stale commands, loses accuracy or must stop, which is a safety risk.
2	CCTV	Live monitoring and any vision guidance become out of step with the real scene; recorded video can be buffered, so the harm is moderate.
3	Office / ERP	Users notice a slower response but the work is not damaged; e-mail and ERP tolerate 100–200 ms.
4	Vibration sensors	One reading per second, so 100 ms is only 10% of the period and the data is used for trends; the harm is negligible.

Alternative Concept Analysis

Suppose the CCTV cameras recorded locally and sent only motion-triggered clips (about 5% of the time) instead of streaming continuously. Recalculate the line traffic, and decide whether the 1 Gbps uplink is still the bottleneck.

Answer

If the 24 cameras send only motion clips (5% of the time), CCTV traffic becomes 24 × 4 Mbps × 0.05 = 4.8 Mbps. The line traffic becomes 32 + 0.24 + 4.8 + 30 = 67.04 Mbps (57.6% less than 158.24 Mbps) and the plant (8 lines) needs 8 × 67.04 = 536.32 Mbps, which is 53.6% of a 1 Gbps uplink, below the 70% planning limit. The 1 Gbps uplink is therefore no longer the bottleneck. The largest source is now the robot controllers (32 Mbps), and short bursts of clips when many cameras trigger together still need a priority queue so that robot commands are not delayed.

TASK 2 – Count the Domains in the Contractor's Drawing 3 Marks

Bloom's Level: K5 – Evaluate

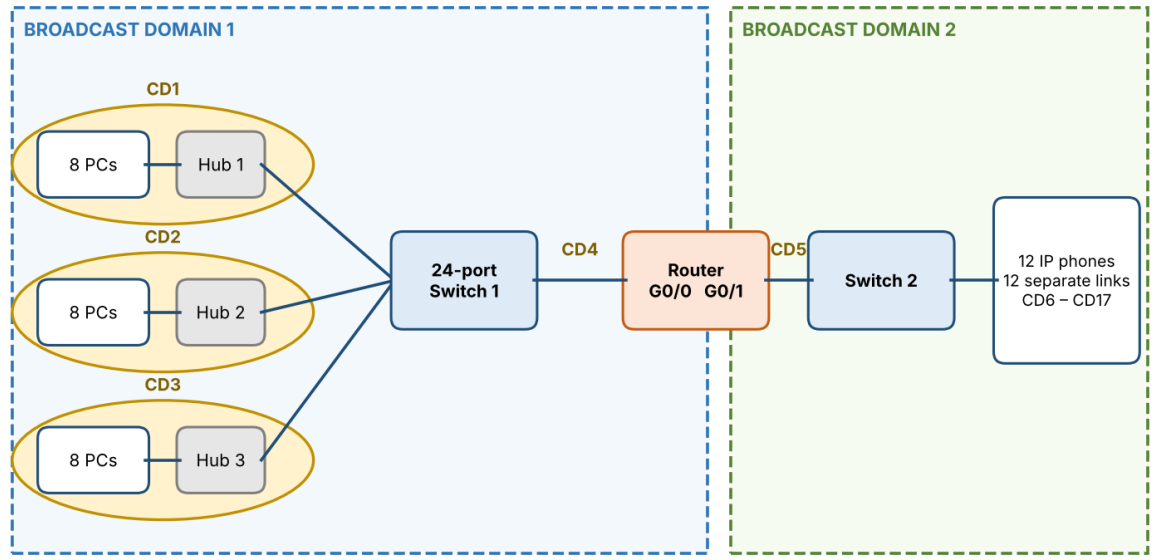
The contractor's drawing for the office block shows: three hubs with 8 PCs each, all three hubs connected to one 24-port switch, the switch connected to router interface G0/0, and router interface G0/1 connected to a second switch with 12 IP phones.

Work out:

- The number of collision domains and broadcast domains, with a marked drawing
- The OSI layer at which each device works, and why a hub cannot stop collisions
- Which connections are point-to-point and which are multipoint
- Which standard and which body defines each of: Ethernet (IEEE 802.3), Wi-Fi (IEEE 802.11), IPv4 (IETF RFC 791), the OSI model (ISO 7498), and 5G (3GPP/ITU) – and why the plant should buy only standard-compliant equipment

Answer

1) Collision and broadcast domains.



Hub = 1 shared collision domain (CD1–CD3); every switch/router link = its own collision domain; router splits the two broadcast domains. Total: 17 CD, 2 BD.

Part of the drawing	Collision domains	Broadcast domains
Hub 1 with its 8 PCs and its switch port	1 (CD1)	Domain 1

Part of the drawing	Collision domains	Broadcast domains
Hub 2 with its 8 PCs and its switch port	1 (CD2)	Domain 1
Hub 3 with its 8 PCs and its switch port	1 (CD3)	Domain 1
Switch 1 – router G0/0 link	1 (CD4)	Domain 1
Router G0/1 – switch 2 link	1 (CD5)	Domain 2
12 IP phone links to switch 2	12 (CD6–CD17)	Domain 2
Total	**17 collision domains**	**2 broadcast domains**

A hub makes all its ports one collision domain, each switch or router port is its own collision domain, and only the router (G0/0 and G0/1) separates the broadcast domains.

2) OSI layer of each device. Hub: layer 1 (physical). Switch: layer 2 (data link). Router: layer 3 (network). PCs and IP phones: all layers up to layer 7 (end devices). A hub cannot stop collisions because it only repeats the electrical signal to every other port; it does not read addresses or store frames, so all attached devices share one half-duplex medium, and two stations that transmit at the same time collide (CSMA/CD).

3) Point-to-point and multipoint. Point-to-point: switch 1 – router G0/0, router G0/1 – switch 2, each IP phone – switch 2, each hub – switch 1 uplink, and each PC – hub cable (as a physical cable). Multipoint (shared medium): each hub segment, where the 8 PCs and the uplink share one half-duplex collision domain.

4) Standards and bodies.

Standard	Body	What it defines
Ethernet (IEEE 802.3)	IEEE	wired LAN physical layer and MAC
Wi-Fi (IEEE 802.11)	IEEE (certified by the Wi-Fi Alliance)	wireless LAN
IPv4 (RFC 791)	IETF	IP addressing and packet forwarding
OSI model (ISO 7498)	ISO	seven-layer reference model
5G	3GPP (specifications) and ITU (IMT-2020 requirements)	mobile radio and core network

The plant should buy only standard-compliant equipment because devices from different vendors then interoperate, the plant is not locked in to one supplier, equipment can be replaced or expanded when the network grows to five plants, products are tested and certified, and competition keeps the cost low.

Alternative Concept Analysis

A colleague proposes replacing the three hubs with three cheap unmanaged switches. Recount the collision and broadcast domains, and state what this change fixes and what it does not.

Answer

With three unmanaged switches in place of the hubs, every PC has its own link: 24 PC links + 3 uplinks to switch 1 + 1 (switch 1 – router) + 1 (router – switch 2) + 12 phone links = 41 collision domains. The broadcast domains remain 2, because switches do not separate broadcasts.

What it fixes: collisions disappear (full duplex), each PC gets a dedicated link instead of sharing one hub with 7 others, and CSMA/CD waiting is removed. **What it does not fix:** the 24 PCs are still one broadcast domain (ARP/broadcast storms), there are no VLANs or security separation, no QoS or monitoring (unmanaged), no loop protection, and the three uplinks to switch 1 are still shared bottlenecks.

TASK 3 – Read a Captured Frame Layer by Layer 4 Marks

Bloom's Level: K5 – Evaluate

A capture of one robot command shows the following fields:

```
Ethernet II  Dst: 00:1d:9c:c7:b0:41   Src: 3c:52:82:4e:11:07   Type: 0x0800
IPv4         Src: 10.10.1.25  Dst: 10.10.9.4  TTL: 61  Protocol: 6 (TCP)
TCP          Src Port: 51522  Dst Port: 502 (Modbus)  Seq: 1201  Ack: 88  Len: 12
Data         12 bytes (Modbus write-register command)
```

Work out:

- The OSI layer and the protocol data unit (frame, packet, segment, data) of each line
- How many routers the packet crossed if the controller sets TTL to 64
- The total bytes on the wire (include the 4-byte FCS) and the percentage that is overhead; then repeat for a 1,400-byte camera packet – what does this tell you about small commands?
- Which layer is responsible when (a) the MAC address is wrong, (b) the TTL reaches 0, (c) the destination port is closed
- How the TCP/IP model maps onto these seven layers

Answer

1) Layer and PDU of each line.

Line	OSI layer	Protocol data unit
Ethernet II	2 – Data link	Frame
IPv4	3 – Network	Packet
TCP	4 – Transport	Segment
Data (Modbus write-register command)	7 – Application	Data (message)

2) Routers crossed. The controller sets TTL 64 and the captured TTL is 61, so the packet crossed 64 − 61 = 3 routers (each router decrements TTL by 1).

3) Bytes on the wire and overhead. Robot command: Ethernet header 14 + IPv4 20 + TCP 20 + data 12 + FCS 4 = 70 bytes; overhead = 58 bytes = 58/70 = 82.9%. Camera packet with 1,400 bytes of data: 14 + 20 + 20 + 1,400 + 4 = 1,458 bytes; overhead = 58/1,458 = 4.0%. The headers are the same 58 bytes in both cases, so a small 12-byte command wastes about 83% of the bandwidth while a large camera packet wastes only 4%: small commands are very inefficient in bytes (so many small commands must be given priority and not mixed with bulk traffic).

4) Responsible layer. (a) Wrong MAC address: data link layer (the frame is not accepted by the destination NIC). (b) TTL reaches 0: network layer (the router discards the packet and sends an ICMP Time Exceeded message). (c) Destination port closed: transport layer (TCP replies with a reset, RST).

5) TCP/IP to OSI mapping.

TCP/IP layer	OSI layers
Application	Application (7), Presentation (6), Session (5)
Transport	Transport (4)
Internet	Network (3)
Network access (link)	Data link (2) and Physical (1)

Complete the Code

```
"""How data is wrapped by each layer at the sender and unwrapped at the receiver."""

LAYERS = [("Application", "HTTP"), ("Transport", "TCP"), ("Network", "IP"),
          ("Data link", "ETH")]



def send(message):
pdu = message
    for layer, tag in LAYERS:
        # TODO 1: Wrap pdu with this layer's header "[tag|" and, for the data link layer
        #   only, also a trailer "|FCS]"; for other layers close with "]". Print the
        #   layer name and the size of the new PDU
pass  # replace this line with your code
    return pdu



def receive(frame):
pdu = frame
    for layer, tag in reversed(LAYERS):
        head = "[" + tag + "|"
        tail = "|FCS]" if tag == "ETH" else "]"
        assert pdu.startswith(head) and pdu.endswith(tail), f"{layer}: bad header"
pdu = pdu[len(head):-len(tail)]
    return pdu



frame = send("GET /results")
print("on the wire:", frame)
print("delivered  :", receive(frame))
print(f"overhead  : {len(frame) - len('GET /results')} bytes for 12 bytes of data")
```

Expected output:

```
Application  adds HTTP ->  19 bytes
Transport    adds TCP  ->  25 bytes
Network      adds IP   ->  30 bytes
Data link    adds ETH  ->  40 bytes
on the wire: [ETH|[IP|[TCP|[HTTP|GET /results]]]|FCS]
delivered  : GET /results
overhead   : 28 bytes for 12 bytes of data
```

Change the message to 1,400 bytes and compare the overhead percentage with the 12-byte message.

Answer

Result of the completed program. Each layer wraps the data: HTTP gives 19 bytes, TCP 25 bytes, IP 30 bytes and Ethernet 40 bytes, so the frame on the wire is [ETH|[IP|[TCP|[HTTP|GET /results]]]|FCS], the receiver unwraps it to “GET /results”, and the overhead is 28 bytes for 12 bytes of data – identical to the expected output.

With a 1,400-byte message the overhead is still 28 bytes: 28/1,428 = 1.96%, compared with 28/40 = 70.0% for the 12-byte message. The overhead percentage is about 36 times smaller, so large messages hide the header cost while small messages are dominated by it.

Alternative Concept Analysis

An industrial protocol could carry the robot command directly inside an Ethernet frame (EtherType 0x8892) with no IP or TCP header. Calculate the bytes saved per command and the commands per second this frees on a 100 Mbps link, and identify what the plant loses by giving up IP.

Answer

A raw Ethernet frame (EtherType 0x8892) needs 14 + 12 + 4 = 30 bytes, i.e. 40 bytes (20 IP + 20 TCP) fewer than 70 bytes. However, Ethernet has a minimum frame size of 64 bytes, so the 12-byte command is padded and the real frame is 64 bytes: the actual saving is 70 − 64 = 6 bytes per command (8.6%).

On a 100 Mbps link: with IP/TCP 100,000,000 / (70 × 8) = 178,571 commands/s; with raw Ethernet 100,000,000 / (64 × 8) = 195,313 commands/s. About 16,741 more commands per second (+9.4%) are possible. (Counting the 8-byte preamble and 12-byte gap, the figures are 138,889 and 148,810 commands/s, a gain of 9,921 per second.)

What is lost by giving up IP: routing – the frame cannot cross a router, so all devices must stay in one layer-2 network and cannot reach other plants or the head office; no TCP reliability (retransmission, ordering, flow control), which the protocol must provide itself; no IP addressing, standard IP security and ordinary network tools; and growth to five plants becomes difficult.

Evidence Expected

A layer-by-layer reading of the capture, correct overhead calculations and a fault-to-layer mapping.

TASK 4 – Choose a Medium for Each Run (There Is a Trap) — 4 Marks

Bloom's Level: K5 – Evaluate

The plant has four cable runs: (a) a robot cell next to arc-welding machines, 85 m; (b) the office floor to the shop-floor switch, 140 m; (c) the shop floor to the data centre, 550 m; (d) the plant to the head office, 400 km.

Work out:

- A medium for each run (Cat6 UTP, Cat6A S/FTP, multimode fibre, single-mode fibre, leased line or wireless) with the deciding reason
- Which run cannot use copper Ethernet at all, and why (look at the distance limits)
- The Shannon capacity of a 250 MHz cable with an SNR of 40 dB and of the same cable near the welders with an SNR of 15 dB – what does the difference mean for the robot cell?
- Why the speed measured on a link is always below the Shannon limit

Answer

1) Medium for each run.

Run	Medium	Deciding reason
(a) Robot cell, 85 m, next to arc welders	Cat6A S/FTP (shielded, grounded)	85 m is within the 100 m copper limit and the foil-and-braid shield resists welding noise (multimode fibre, which is immune to noise, if shielding is not enough)
(b) Office to shop-floor switch, 140 m	Multimode fibre (OM3/OM4)	140 m is above the 100 m limit of Cat6/Cat6A – the trap; fibre has no distance problem here
(c) Shop floor to data centre, 550 m	Single-mode fibre (OS2)	Far above 100 m; multimode reaches 550 m only at 1 Gbps and only 300–400 m at 10 Gbps, single-mode gives headroom for 10 Gbps and later
(d) Plant to head office, 400 km	Leased line (carrier single-mode fibre with amplifiers)	No plant-owned cable can span 400 km; the carrier provides the long-haul fibre

2) Which run cannot use copper Ethernet. Copper Ethernet (Cat6/Cat6A) is limited to 100 m, so runs (b) 140 m, (c) 550 m and (d) 400 km cannot use it. Run (b) is the trap because 140 m looks like an ordinary office cable. Run (d) cannot be built with any plant-owned cable and needs a carrier link.

3) Shannon capacity of the 250 MHz cable. SNR 40 dB = 10,000; C = 250 MHz × $\log_2(1 + 10{,}000)$ = 250 × 13.29 = **3,322 Mbps (3.32 Gbps)**. SNR 15 dB = 31.62; C = 250 MHz × $\log_2(32.62)$ = 250 × 5.03 = **1,257 Mbps (1.26 Gbps)**. Capacity falls by 62% near the welders. The robot cell needs only about 0.8 Mbps per robot (32 Mbps for 40), so capacity itself is not the problem; the lower SNR raises the bit-error rate, giving corrupted frames, CRC errors and retransmissions that can push a command beyond the 10 ms deadline, so the welding area needs shielded cable or fibre.

4) Why the measured speed is below the Shannon limit. Shannon's value is the theoretical maximum with ideal coding and unlimited complexity. A real link uses finite modulation and coding, loses capacity to headers, preamble and gaps, ACKs, retransmissions and contention, suffers real noise, crosstalk and attenuation, and is limited by hardware (NIC, CPU, disk) – so the measured rate is always lower.

Complete the Code

Complete TODO 1 and TODO 2 below, and run the program with Python 3.

```
"""Nyquist and Shannon limits for the transmission media proposed for a campus."""
import math


def nyquist(bandwidth_hz, levels):
    # TODO 1: Return the Nyquist maximum bit rate for a noiseless channel: 2 x bandwidth
    #   x log2(levels)
pass  # replace this line with your code


def shannon(bandwidth_hz, snr_db):
    # TODO 2: Convert the SNR from decibels to a ratio (10 ** (dB / 10)) and return the
    #   Shannon capacity: bandwidth x log2(1 + SNR)
pass  # replace this line with your code


media = [("Telephone line", 3_100, 30), ("Cat6 pair", 250e6, 40),
         ("Wi-Fi 6 channel", 160e6, 25)]
for name, bw, snr in media:
    print(f"{name:<16} Shannon limit {shannon(bw, snr) / 1e6:10.2f} Mbps")
print(f"Nyquist, 3,100 Hz with 4 levels: {nyquist(3_100, 4) / 1e3:.1f} kbps")
```

Expected output:

```
Telephone line   Shannon limit       0.03 Mbps
Cat6 pair        Shannon limit    3321.96 Mbps
Wi-Fi 6 channel  Shannon limit    1329.50 Mbps
Nyquist, 3,100 Hz with 4 levels: 12.4 kbps
```

Find the SNR (in dB) a 3,100 Hz telephone line would need to carry 56 kbps, and judge whether that is realistic.

Answer

Result of the completed program.

Medium / case	Result
Telephone line (3,100 Hz, 30 dB)	Shannon limit 0.03 Mbps (30.9 kbps)
Cat6 pair (250 MHz, 40 dB)	Shannon limit 3,321.96 Mbps
Wi-Fi 6 channel (160 MHz, 25 dB)	Shannon limit 1,329.50 Mbps
Nyquist, 3,100 Hz with 4 levels	12.4 kbps

These agree with the expected output. **SNR needed for 56 kbps:** $\log_2(1 + S) = 56{,}000 / 3{,}100 = 18.06$, so S = 2^18.06 − 1 ≈ 274,000, i.e. $10 \cdot \log_{10}(274{,}000)$ = **54.4 dB**. This is not realistic: an analogue telephone line usually gives only about 30–35 dB (Shannon limit about 31–36 kbps); 56 kbps is reached only because modems such as V.90 use a digital connection on the other side.

Alternative Concept Analysis

What if the robot cell used Wi-Fi 6 instead of shielded cable? Using the welding-area SNR of 15 dB on a 160 MHz channel, calculate the Shannon limit, and decide whether a wireless link is acceptable for 10 ms robot commands.

Answer

With an SNR of 15 dB on a 160 MHz Wi-Fi 6 channel, C = 160 MHz × $\log_2(1 + 31.62)$ = 160 × 5.03 = **804 Mbps**. The capacity is far more than the 32 Mbps the robots need, but capacity is not the issue: Wi-Fi uses CSMA/CA with random back-off, retransmissions and interference from the welders, so it cannot guarantee a 10 ms delay for every command. A wireless link is therefore not acceptable as the only connection for the 10 ms robot commands; fixed robots should stay on shielded cable or fibre, and Wi-Fi 6 should be used only for non-critical traffic.

TASK 5 – Wi-Fi 6 or 5G for Moving Equipment? 4 Marks

Bloom's Level: K5 – Evaluate

An AGV travels at 2 m/s along a 1.2 km route. Wi-Fi access points are 30 m apart, and a Wi-Fi hand-over takes about 150 ms. A private 5G cell covers 300 m, and a 5G hand-over takes about 20 ms. AGV commands have a 10 ms deadline.

Work out:

- The number of hand-overs per trip for Wi-Fi and for 5G, and the total time per trip during which the AGV is not reachable
- Which of the three 5G service types (eMBB, URLLC, mMTC) the AGVs, the cameras and the sensors each need
- Where Bluetooth, Zigbee or LoRa would be a better choice than Wi-Fi or 5G in the plant
- Whether Wi-Fi 6 with fast roaming (802.11r) could meet the deadline – and what remains risky

Answer

1) Hand-overs and unreachable time. The trip lasts 1,200 m / 2 m/s = 600 s.

Technology	Cells on the route	Hand-overs	Time not reachable per trip
Wi-Fi (AP every 30 m)	1,200 / 30 = 40	39	39 × 150 ms = 5.85 s (0.98% of the trip)
Private 5G (300 m cell)	1,200 / 300 = 4	3	3 × 20 ms = 0.06 s (0.01% of the trip)

Each 150 ms Wi-Fi hand-over spans 15 command deadlines of 10 ms and the AGV travels 0.3 m without commands; a 5G hand-over spans 2 deadlines and 0.04 m.

2) 5G service types. AGVs: URLLC (ultra-reliable low latency, 10 ms deadline). CCTV cameras: eMBB (4 Mbps continuous video each, high bandwidth). Sensors: mMTC (very many low-rate devices, 50 B/s each).

3) Where Bluetooth, Zigbee or LoRa are better. Bluetooth (low energy): short-range links such as handheld scanners, headsets, tool and asset tags. Zigbee: low-power mesh of fixed sensors and actuators inside a cell or building (environment, lighting, switches). LoRa: long-range (kilometres), very

low-rate, battery-powered sensors such as outdoor yard, tank or remote-building sensors that report once per minute or less. All of them need less power and cost less than Wi-Fi or 5G for such devices.

4) Wi-Fi 6 with 802.11r. Fast roaming reduces a hand-over from about 150 ms to roughly 50 ms or less, so 39 hand-overs still cost about 39 × 50 ms = 1.95 s per trip and each hand-over still spans several 10 ms deadlines – it does not meet the deadline for every command. What remains risky: the roaming time is variable and not guaranteed, access contention (CSMA/CA) and retransmissions add jitter, interference from welding equipment, and no deterministic latency.

Alternative Concept Analysis

The sensors could use LoRaWAN (a few kbps, kilometres of range, years on a battery) instead of Wi-Fi. Calculate whether 600 sensors at 50 bytes per second fit within one LoRa gateway's capacity of roughly 5 kbps, and decide which sensors should stay on Wi-Fi.

Answer

600 sensors × 50 B/s × 8 = 240,000 bit/s = 240 kbps, which is 240/5 = 48 times the capacity of one LoRa gateway (about 5 kbps); one gateway can serve only 5,000 / 400 = 12.5 sensors at that rate, so 48 gateways would be needed – this does not fit. LoRa can be used only for sensors that send much less often: if each sensor sends 50 bytes once a minute, 600 sensors need 600 × 50 × 8 / 60 = 4 kbps (80% of one gateway), which fits (two gateways give margin). Decision: the vibration sensors that report every second should stay on Wi-Fi (or cable), while slow sensors such as temperature and environment, reporting once a minute or less, can move to LoRaWAN to save power and Wi-Fi capacity.

TASK 6 – Interpret an Ethernet Port Report **4 Marks**

Bloom's Level: K5 – Evaluate

A shop-floor PC is slow. Its network interface reports:

```
Speed: 100Mb/s   Duplex: Half   Auto-negotiation: off
Link detected: yes   rx_errors: 2381   late_collisions: 412   crc_errors: 1907
```

Work out:

- What a late collision and a CRC error each mean, and what the combination suggests about the two ends of the link
- The likely cause (a duplex mismatch with a gigabit switch port set to auto) and the exact configuration change on Linux (ethtool) or Windows
- Whether the MAC address 01:00:5e:0a:01:02 belongs to a single device or a group, and what the first three bytes of 3c:52:82:4e:11:07 identify
- Whether jumbo frames (9,000 bytes) would help the CCTV storage server, and what must also change for them to work

Answer

1) Meaning of the counters. A late collision is a collision detected after the first 64 bytes (512 bit times) of a frame have been sent; on a correct network it cannot happen, so it indicates a duplex mismatch or a cable that is too long. A CRC error means the frame check sequence does not match: the frame was damaged or cut short. Together they show that the two ends of the link do not agree: one end works half duplex (CSMA/CD) while the other transmits as full duplex, so frames collide and are cut off or corrupted.

2) Cause and fix. Likely cause: duplex mismatch – the PC is fixed at 100 Mb/s half duplex with auto-negotiation off while the gigabit switch port works as full duplex/auto. Fix: re-enable auto-negotiation on the PC (or set both ends the same, 1 Gbps full duplex). On Linux: ethtool -s eth0 autoneg on (or speed

1000 duplex full autoneg on); on Windows: Device Manager → network adapter → Advanced → Speed & Duplex → Auto Negotiation. Afterwards late collisions and CRC errors should stop increasing.

3) MAC addresses. 01:00:5e:0a:01:02: the first byte 01 has its least significant bit = 1, so it is a group (multicast) address – the prefix 01:00:5e is IPv4 multicast, mapped from group address x.10.1.2 – not a single device. For 3c:52:82:4e:11:07 the first three bytes 3c:52:82 are the OUI (Organisationally Unique Identifier) that identifies the manufacturer of the network card; the last three bytes are the card's serial number (the address is also unicast and globally administered).

4) Jumbo frames. Yes, for the CCTV storage server: header overhead of IP/TCP/Ethernet (58 bytes) falls from 58/1,518 = 3.8% with 1,500-byte frames to 58/9,018 = 0.64% with 9,000-byte frames, and the server handles 6 times fewer frames, so CPU load and interrupts fall and throughput rises. For this to work every device on the path (NIC, switches, storage server) must support and be configured with the same 9,000-byte MTU; otherwise frames are dropped or fragmented. Jumbo frames should be used only on the storage VLAN, because a 9,018-byte frame takes 72 µs to send at 1 Gbps instead of 12 µs and would delay robot traffic.

Alternative Concept Analysis

Compare a shop floor wired as a ring of industrial switches with fast ring recovery against a star of switches. For one cable cut, state the recovery time each needs and which devices lose connection.

Answer

For one cable cut	Ring of industrial switches with fast recovery	Star of switches
Recovery time	10–200 ms (for example MRP; 0 ms with HSR/PRP); RSTP would need 1–6 s	None – no automatic recovery; service returns only after the cable is repaired (hours)
Devices that lose connection	None permanently: traffic is sent around the ring the other way; all traffic across the cut stops briefly during recovery	The device behind a cut access cable; if an uplink between switches is cut, every device on that switch

For robot commands with a 10 ms deadline even the 10–200 ms ring recovery can miss several commands, so the ring needs the fastest recovery profile (or HSR/PRP) and the robots need a safe-stop behaviour. A star is simpler but has a single point of failure on each branch, so the ring is preferable for the shop floor.

TASK 7 – Will the Receiver Notice the Error? 4 Marks

Bloom's Level: K5 – Evaluate

Frames between two shop-floor switches are protected by a CRC with generator 1011 ($x^3 + x + 1$).

Work out:

- The CRC bits for the data 1101011011, by hand (show the division)
- Whether a received frame 1101011011101 is accepted or rejected
- The longest burst error this generator is certain to detect, and an error pattern it cannot detect
- The two-dimensional parity of a 4 × 4 block of your choice, and a 4-bit error pattern that two-dimensional parity misses
- The Internet checksum of the four 16-bit words 4500, 0073, 0000, 4000 (hexadecimal)

Answer

1) CRC of 1101011011 with generator 1011. Append three zeros: 1101011011000 and divide modulo 2 (XOR 1011 whenever the leading bit is 1):

Step	XOR 1011 at bit	Result	Note
start	–	1101011011000	data + 000
1	0	0110011011000	
2	1	0011111011000	
3	2	0001001011000	
4	3	0000010011000	
5	5	0000000101000	
6	7	0000000000100	remainder = 100

The CRC bits are **100** and the transmitted frame is **1101011011100**.

2) Received frame 1101011011101. Dividing it by 1011 gives the remainder 001, which is not zero, so the frame is **rejected** (the last bit was corrupted).

3) Burst errors. A CRC with a degree-3 generator is certain to detect every burst error of length **3 bits or less**. It cannot detect an error pattern that is a multiple of the generator polynomial, for example 1011 ($x^3 + x + 1$) itself: frame 1101011011100 with the last four bits flipped by 1011 becomes 1101011010111 and still divides exactly, so the error goes unnoticed (other examples: 11101, or two bits 7 positions apart, 10000001).

4) Two-dimensional parity. Block of 4 × 4 data bits with even row and column parity:

Row	bit 1	bit 2	bit 3	bit 4	Row parity
1	1	1	0	1	1
2	0	1	1	0	0
3	1	0	1	1	1
4	0	0	1	1	0
Column parity	0	0	1	1	0

A 4-bit pattern that is missed: flip the four bits at the corners of a rectangle, for example row 1/column 1, row 1/column 3, row 3/column 1 and row 3/column 3. Every affected row and column contains two flipped bits, so all parities remain correct and the error is not detected.

5) Internet checksum. 4500 + 0073 + 0000 + 4000 = 8573 (hex, no carry). The one's complement is FFFF − 8573 = **7A8C**.

```
"""CRC error detection for frames sent between two switches."""


def mod2 div(dividend, divisor):
    """Return the remainder of modulo-2 (XOR) division of two bit strings."""
    rem = list(dividend)
    for i in range(len(dividend) - len(divisor) + 1):
        # TODO 1: If the current leading bit rem[i] is 1, XOR the divisor into rem
        #   starting at position i
pass  # replace this line with your code
    return "".join(rem[-(len(divisor) - 1):])


def encode(data, gen):
crc = mod2_div(data + "0" * (len(gen) - 1), gen)
    return data + crc


def check(frame, gen):
    return set(mod2_div(frame, gen)) == {"0"}


GEN = "1011"                                    # x^3 + x + 1
frame = encode("1101011011", GEN)
print("transmitted:", frame)
print("no error   :", check(frame, GEN))
bad = frame[:4] + ("1" if frame[4] == "0" else "0") + frame[5:]
print("1-bit error:", check(bad, GEN))
burst = frame[:2] + "".join("1" if b == "0" else "0" for b in frame[2:5]) + frame[5:]
print("3-bit burst:", check(burst, GEN))
```

Expected output:

```
transmitted: 1101011011100
no error   : True
1-bit error: False
3-bit burst: False
```

Find, with the program, an error pattern that generator 1011 cannot detect, and explain why.

Answer

Result of the completed program. transmitted: 1101011011100; no error: True; 1-bit error: False (detected); 3-bit burst: False (detected) – the same as the expected output.

Undetected pattern found: flipping the bits with the pattern 1011 (the generator itself) gives 1101011010111, and the program reports True (accepted) although the frame is corrupted. The reason is that the error polynomial $x^3 + x + 1$ is divisible by the generator, so the remainder is zero; the same happens for any multiple of the generator such as 11101 or 10000001.

Alternative Concept Analysis

Which errors would a simple even-parity bit catch that the CRC-3 generator 1011 misses, and which would it miss? Give one 2-bit and one 4-bit error pattern for each method and state the result.

Answer

Error pattern	Weight	Even parity bit	CRC-3 (generator 1011)
2-bit: 11 (adjacent bits)	2	Missed (even number)	Detected (not a multiple of 1011)
2-bit: 10000001 (7 positions apart)	2	Missed	Missed ($x^7 + 1$ is a multiple of 1011)

Error pattern	Weight	Even parity bit	CRC-3 (generator 1011)
4-bit: 1111 (burst)	4	Missed	Detected (remainder 100)
4-bit: 11101	4	Missed	Missed (multiple of 1011)
3-bit: 1011	3	Detected (odd number)	Missed (equals the generator)

Even parity catches every odd number of bit errors (including the 3-bit pattern 1011 that the CRC misses) but misses every even number of errors. The CRC-3 catches all bursts up to 3 bits and most other patterns (including 2-bit and 4-bit errors that parity misses) and detects about 87.5% of random errors against 50% for a single parity bit, so the CRC is stronger overall, although each method has patterns that the other catches.

TASK 8 – Correct the Error or Ask Again? 3 Marks

Bloom's Level: K5 – Evaluate

A vibration sensor sends 16-bit readings over a noisy radio link where retransmission takes 40 ms.

Work out:

- The number of Hamming check bits needed for a 16-bit word (use $2^r \geq m + r + 1$) and the overhead percentage
- The minimum Hamming distance of the code {00000, 01011, 10101, 11110}, and how many errors it can detect and correct
- The corrected data when the Hamming(7,4) codeword 0110011 arrives as 0110111
- Whether forward error correction or retransmission (ARQ) suits this link, and why video streaming and 5G use FEC

Answer

1) Hamming check bits for m = 16. $2^r \geq m + r + 1$: r = 4 gives 16 ≥ 21 (false); r = 5 gives 32 ≥ 22 (true), so **5 check bits** are needed (21-bit codeword). Overhead = 5/16 = 31.25% of the data (23.8% of the transmitted bits).

2) Minimum Hamming distance of {00000, 01011, 10101, 11110}. Distances: 00000–01011 = 3, 00000–10101 = 3, 00000–11110 = 4, 01011–10101 = 4, 01011–11110 = 3, 10101–11110 = 3. The minimum distance is **3**: it can detect up to 2 errors and correct 1 error.

3) Hamming(7,4) word received as 0110111. s1 (positions 1,3,5,7) = 0⊕1⊕1⊕1 = 1; s2 (positions 2,3,6,7) = 1⊕1⊕1⊕1 = 0; s3 (positions 4,5,6,7) = 0⊕1⊕1⊕1 = 1. The syndrome is 1 + 0 + 4 = 5, so bit 5 is wrong; flipping it gives the codeword 0110011, and the corrected data (positions 3, 5, 6, 7) is **1011**.

4) FEC or ARQ. Forward error correction suits this noisy radio link: a retransmission takes 40 ms and may itself be corrupted, whereas FEC corrects a single-bit error at once with a constant delay, needs no feedback channel and costs only 5 extra bits per reading. Video streaming and 5G use FEC because waiting for retransmissions would exceed the real-time deadline, one stream may go to many receivers that cannot all send acknowledgements, and round-trip times can be long (5G also adds HARQ for the remaining errors).

```
"""Hamming(7,4) code: correct a single-bit error in a 4-bit sensor reading."""


def encode(d):                      # d = [d1, d2, d3, d4]
    p1 = d[0] ^ d[1] ^ d[3]
    p2 = d[0] ^ d[2] ^ d[3]
    p3 = d[1] ^ d[2] ^ d[3]
    return [p1, p2, d[0], p3, d[1], d[2], d[3]]      # positions 1..7


def decode(c):
    # TODO 1: Compute the syndrome bits s1 (positions 1,3,5,7), s2 (2,3,6,7) and s3
    #    (4,5,6,7); pos = s1 + 2*s2 + 4*s3. If pos is not 0, flip bit pos (index pos-1).
    #    Return pos and the data bits at positions 3, 5, 6, 7
pass  # replace this line with your code


word = encode([1, 0, 1, 1])
print("codeword       :", "".join(map(str, word)))
for flip in (None, 5):
rx = word[:]
    if flip:
rx[flip - 1] ^= 1
pos, data = decode(rx)
print(f"error at {str(flip):<4}: syndrome {pos}, data {''.join(map(str, data))}")
```

Expected output:

```
codeword       : 0110011
error at None: syndrome 0, data 1011
error at 5   : syndrome 5, data 1011
```

Flip two bits of the codeword and show that the decoder now “corrects” the wrong bit.

Answer

Result of the completed program. codeword: 0110011; no error: syndrome 0, data 1011; error at bit 5: syndrome 5, data 1011 (the error is corrected) – the same as the expected output.

Two flipped bits. Flip bits 3 and 5 of 0110011: the received word is 0100111, the syndrome becomes 6, so the decoder “corrects” bit 6, which was correct, and returns the data 0101 instead of 1011 without any warning. With minimum distance 3 the code can correct only one error; two errors look like a single error at another position.

Alternative Concept Analysis

At a bit-error rate of 1 in 1,000, compare sending 16-bit readings with Hamming correction against retransmitting on error: calculate the bits sent per reading and the average delay for each, using the 40 ms retransmission time.

Answer

At BER = 10^{-3}	Hamming correction (21,16)	Retransmission (ARQ)
Bits sent per reading	21	16 × 1.0161 = 16.26 (average)
Probability that action is needed	≥ 2 bit errors in 21 bits = 2.07×10^{-4}	≥ 1 bit error in 16 bits = 1.59%
Average extra delay	$2.07 \times 10^{-4} \times 40$ ms = 0.008 ms (only if uncorrectable)	0.0161 × 40 ms = 0.645 ms
Worst-case delay	0 ms for 99.98% of readings (40 ms for 0.02%)	40 ms or more for 1.6% of readings

ARQ sends 22.6% fewer bits (16.3 against 21) but its average delay is about 78 times higher and about 1.6% of the readings arrive 40 ms late, which gives jitter. For time-sensitive vibration data the Hamming-

corrected transmission is the better choice; ARQ is preferable only when bandwidth is more important than delay.

TASK 9 – How Big Must the Window Be? 4 Marks

Bloom's Level: K5 – Evaluate

The head-office link is 400 km of fibre at 1 Gbps; frames are 1,500 bytes; light travels at 2×10^8 m/s in fibre.

Work out:

- The transmission time, the propagation time and the ratio a = propagation / transmission
- The link utilisation with stop-and-wait ARQ – and what it means in Mbps
- The smallest window that keeps the link fully used, and the number of sequence-number bits needed for Go-Back-N and for Selective Repeat with that window
- Which of the two ARQ protocols resends fewer frames after one loss, and what it costs the receiver
- How HDLC or PPP frames mark the start and end of a frame

Answer

1) Timing. Frame = 1,500 B = 12,000 bits. Transmission time = 12,000 / 10^9 = **12 µs**. Propagation time = 400,000 m / (2×10^8 m/s) = **2 ms**. a = 2,000 µs / 12 µs = **166.67**.

2) Stop-and-wait utilisation = 1/(1 + 2a) = 1/334.33 = **0.299%**, i.e. about 2.99 Mbps out of 1,000 Mbps – the 1 Gbps link is almost completely idle.

3) Window and sequence numbers. The smallest window that keeps the link fully used is W = 1 + 2a = 334.33, i.e. **335 frames**. Go-Back-N needs $W \le 2^k - 1$: $2^8 - 1 = 255$ is too small and $2^9 - 1 = 511$ is enough, so **9 bits**. Selective Repeat needs $W \le 2^{k-1}$: $2^9 = 512 \ge 335$, so **10 bits**.

4) Frames resent after one loss. Selective Repeat resends only the lost frame, whereas Go-Back-N resends the lost frame and all following frames (up to 335). The cost for the receiver is a buffer for out-of-order frames – up to 335 × 1,500 B ≈ 502 kB – and more complex logic (selective acknowledgements and re-ordering); a Go-Back-N receiver needs no buffer.

5) Frame delimiting. HDLC marks the start and end of a frame with the flag 01111110 (0x7E) and uses bit stuffing (a 0 is inserted after five consecutive 1s in the data). PPP also uses the flag 0x7E, with byte stuffing: a 0x7E or 0x7D in the data is replaced by 0x7D followed by the byte XOR 0x20.

```
"""Link utilisation of stop-and-wait and sliding-window protocols on different links."""



def utilisation(frame_bits, rate_bps, distance_m, window, speed=2e8):
tx = frame_bits / rate_bps                  # transmission time
    prop = distance_m / speed                   # propagation time
    # TODO 1: a = prop / tx. Utilisation is window / (1 + 2a), but never more than 1.
    #   Return the utilisation and a
pass  # replace this line with your code



links = [("Campus LAN, 200 m, 1 Gbps", 1e9, 200),
         ("City fibre, 50 km, 1 Gbps", 1e9, 50_000),
         ("Satellite, 36,000 km, 10 Mbps", 10e6, 36_000_000)]
for name, rate, dist in links:
    u1, a = utilisation(12_000, rate, dist, 1)
    u7, _ = utilisation(12_000, rate, dist, 7)
    u127, _ = utilisation(12_000, rate, dist, 127)
    print(f"{name:<31} a = {a:7.2f}  stop-and-wait {u1:6.1%}  "
f"W=7 {u7:6.1%}  W=127 {u127:6.1%}")
```

Expected output:

```
Campus LAN, 200 m, 1 Gbps       a =    0.08  stop-and-wait  85.7%  W=7 100.0%  W=127 100.0%
City fibre, 50 km, 1 Gbps       a =   20.83  stop-and-wait   2.3%  W=7  16.4%  W=127 100.0%
Satellite, 36,000 km, 10 Mbps   a =  150.00  stop-and-wait   0.3%  W=7   2.3%  W=127  42.2%
```

Calculate the smallest window that gives 100% utilisation on the satellite link.

Answer

Result of the completed program.

Link	a	Stop-and-wait	W = 7	W = 127
Campus LAN, 200 m, 1 Gbps	0.08	85.7%	100.0%	100.0%
City fibre, 50 km, 1 Gbps	20.83	2.3%	16.4%	100.0%
Satellite, 36,000 km, 10 Mbps	150.00	0.3%	2.3%	42.2%

These equal the expected output. **Smallest window for 100% on the satellite link:** a = 150, so W = 1 + 2a = **301 frames**.

Alternative Concept Analysis

Suppose the head-office link used Selective Repeat with a window of 64 instead of the window you calculated. Calculate the resulting utilisation and throughput, and the receiver buffer space needed.

Answer

With Selective Repeat and a window of 64 on the head-office link (1 + 2a = 334.33): utilisation = 64/334.33 = **19.1%**, throughput = 0.191 × 1,000 Mbps ≈ **191 Mbps**, so about 81% of the link stays idle. The receiver needs a buffer for 64 frames = 64 × 1,500 B = **96,000 bytes (96 kB)**, compared with about 502 kB for the window of 335, and 7 sequence-number bits are enough instead of 10. The smaller window saves buffer memory but wastes most of the 1 Gbps link.

TASK 10 – Find Where the Delay Comes From — 4 Marks

Bloom's Level: K5 – Evaluate

A traceroute from the plant to the head-office server gives:

```
1  10.10.1.1        0.6 ms   0.5 ms   0.6 ms
2  100.64.0.1       3.1 ms   2.9 ms   3.4 ms
3  172.31.4.9       4.0 ms  38.2ms    4.2 ms
4  103.21.8.17     11.8 ms  11.5ms   12.1ms
5  * * *
6  10.200.0.10     12.6 ms  12.4ms   12.9ms
```

Work out:

- The propagation delay you would expect for 400 km of fibre, and how much of the measured delay it explains
- Why hop 3 shows one value of 38.2 ms, and why hop 5 shows only stars – are they faults?
- Which hop adds the most delay
- Your own traceroute, ping and Wireshark capture of a real connection, with the headers of each layer identified

Answer

1) Propagation delay. 400 km / (2 × 10^8 m/s) = **2 ms** one way, 4 ms round trip. The traceroute times are round-trip times (12.6 ms at hop 6), so propagation explains 4/12.6 = about **32%**; the remaining 8.6 ms comes from a fibre route longer than 400 km, queuing, processing and serialisation at the routers.

2) Hop 3 and hop 5. Hop 3 shows 38.2 ms in only one of three probes, while the other probes (4.0 and 4.2 ms) and all later hops (11.8 ms and above) are normal, so the data path was not delayed: the router answered that single ICMP probe late (busy control plane or ICMP rate limiting). Hop 5 shows only stars because that router does not return ICMP Time Exceeded messages (filtered or rate-limited), but it forwards traffic normally because hop 6 answers. Neither is a fault.

3) Hop with the most delay. Average increase per hop (taking the normal values for hop 3): hop 1 0.6 ms, hop 2 +2.6 ms, hop 3 +1.0 ms, hop 4 (103.21.8.17) **+7.7 ms** (4.1 → 11.8 ms), hop 6 +0.8 ms. Hop 4 adds the most delay – it is the long-distance provider segment.

4) Own measurement. This part must be your own capture. Run ping and traceroute to the head-office server and capture them in Wireshark. In each packet identify: Ethernet II header (14 bytes: destination and source MAC, type 0x0800), IPv4 header (20 bytes: source and destination address, TTL, protocol 1 = ICMP or 6 = TCP), then the ICMP echo (type 8/0) or TCP header (ports, sequence and acknowledgement numbers), and the data; in the traceroute replies, find the ICMP Time Exceeded (type 11) packets.

Alternative Concept Analysis

Compare continuous monitoring with SNMP counters and flow records (NetFlow) against occasional ping and traceroute tests: which of the problems in the traceroute above would each have detected, and how quickly?

Answer

Problem in the traceroute	Occasional ping/traceroute	Continuous SNMP counters + NetFlow
Hop 3 spike (38.2 ms)	Seen only if the test happens to run; one sample, so it looks like a fault (false alarm)	Interface utilisation and drops show no congestion, so it is correctly seen as harmless; visible within the polling interval (1–5 min)
Hop 5 stars	Shown immediately when run (looks like a fault, but is not)	Counters show the interface up and forwarding traffic, so no alarm is raised
Hop 4 extra delay (+7.7 ms)	Shows the delay step only when someone runs the test (minutes to hours later)	Utilisation, errors and queue drops reveal congestion within 1–5 min, and NetFlow shows which flows (for example CCTV) fill the link within about a minute

Continuous monitoring finds real problems faster and with fewer false alarms, and keeps history; ping and traceroute are cheap, give an instant check of delay and path, but only when they are run, and can mislead (spikes, stars).

TASK 11 – Defend the Network Design — 4 Marks

Bloom's Level: K5 – Evaluate

Recommend the network design for the plant.

Work out:

- The medium, wireless technology and Ethernet settings for each area
- The error-control and flow-control method for each link, with the calculation that justifies it
- The two biggest risks that remain in your design

- Whether your design still works for five plants

Answer

1) Medium, wireless technology and Ethernet settings.

Area	Medium	Wireless	Ethernet settings
Robot cell (welding), 85 m	Cat6A S/FTP, shielded	None (10 ms commands)	1 Gbps full duplex, auto-negotiation on, own VLAN, strict-priority queue
Office to shop floor, 140 m	Multimode fibre (above the 100 m copper limit)	Wi-Fi 6 with 802.11r (30 Mbps per line)	1 Gbps full duplex, switches only, no hubs (41 collision domains)
Shop floor to data centre, 550 m	Single-mode fibre	–	10 Gbps uplink (plant needs 1,265.9 Mbps; 536.3 Mbps with local CCTV recording)
Plant to head office, 400 km	Leased line / carrier fibre, at least 2 × 1 Gbps	–	Routed link, jumbo frames only on the storage VLAN
Sensors and AGVs	–	LoRa/Zigbee for slow sensors; Wi-Fi for 1 s vibration data; private 5G URLLC for AGVs (3 hand-overs, 0.06 s)	QoS marking by traffic class

2) Error and flow control. Wired LAN links: CRC-32 error detection with TCP retransmission, because wired BER is low and the delay is small. Radio sensor link: Hamming forward correction (5 check bits for 16, 31.25% overhead), because ARQ would add 0.645 ms on average and 40 ms to 1.6% of the readings. Head-office link: sliding window (Selective Repeat) with a window of 335 frames and 10-bit sequence numbers (a = 166.67); stop-and-wait would give only 0.30% = 3 Mbps; the TCP window must cover about 1 Gbps × 4 ms = 500 kB.

3) Two biggest remaining risks. (a) Uplink/WAN capacity and a single point of failure: 8 lines need 1,265.92 Mbps (more than 1 Gbps), and one cable cut would isolate the plant, so a 10 Gbps uplink and a redundant link are required. (b) Radio reliability for moving equipment and welding noise: a 5G hand-over of 20 ms still exceeds the 10 ms deadline (about 2 commands), and noise at 15 dB lowers cable capacity by 62%, so AGVs need safe-stop behaviour and overlapping coverage.

4) Five plants. The design still works if each plant keeps its own VLANs, subnets and local CCTV recording (536 Mbps per plant, not 1,266), uses a 10 Gbps core and its own redundant head-office link, and the 7,500 devices are routed (not one flat network). The window of 335 frames is the same for each plant link, so each link still reaches full utilisation.

Alternative Concept Analysis

Suppose the whole plant used private 5G instead of a wired backbone with Wi-Fi. Using your traffic figures from Task 1, decide which traffic 5G should carry and which must stay on cable, and justify the split.

Answer

Per line the traffic is: robot commands 32 Mbps, vibration sensors 0.24 Mbps, CCTV 96 Mbps, office 30 Mbps. **5G should carry** the robot commands and the sensors (32.24 Mbps per line = 257.9 Mbps for 8 lines): small, delay-critical packets that need URLLC for mobile robots, AGVs and mMTC for many sensors. **Cable must carry** the CCTV (96 Mbps per line, 768 Mbps for 8 lines) and the office traffic (30 Mbps, 240 Mbps for 8 lines): continuous uplink-heavy video would use most of a private 5G cell's uplink capacity, and the office traffic is cheaper and easier on Wi-Fi/cable. The backbone that aggregates all of it (1,008 Mbps of CCTV and office traffic plus the robot traffic = 1,266 Mbps) and the 400 km head-office link must also stay on fibre and leased line because of their capacity and reliability.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design 8 Marks

Bloom's Level: K5 – Evaluate

Changed Requirement

The company adds a **private 5G network** on the shop floor and **50 AGVs** that move parts between lines and must never miss a command.

Re-evaluate your original design. You may **RETAIN** → **MODIFY** → **or REDESIGN** your original solution.

A. Impact Analysis – 2 Marks

Identify which media, wireless choices and error-control settings are affected, and recalculate the traffic from Task 1 with the AGVs added.

Answer

Media: 5G radio units and cells need fibre/PoE cabling, and the backbone must be upgraded from 1 Gbps (it is already 126.6% loaded). **Wireless choice:** Wi-Fi 6 cannot serve moving AGVs, so AGVs move to private 5G URLLC; offices stay on Wi-Fi 6. **Error-control settings:** the radio link now needs 5G FEC/HARQ and hand-over protection, while CRC on wired links and the 335-frame window on the WAN remain.

Recalculated traffic (assumption: each AGV exchanges a 200-byte command every 10 ms = 0.16 Mbps and has one 4 Mbps camera):

Case	Added traffic	Plant total (8 lines)
Task 1 original	–	1,265.92 Mbps
+ 50 AGV commands (50 × 0.16)	8 Mbps	1,273.92 Mbps
+ 50 AGV cameras live (50 × 4)	200 Mbps	1,473.92 Mbps (147% of 1 Gbps)
With local CCTV recording (task 1 alternative) + AGVs	8 + 10 Mbps	554.32 Mbps

B. Test the Original Design – 2 Marks

Show, using your hand-over and delay calculations, whether the original design can serve 50 moving AGVs.

Answer

The AGV route is 1.2 km at 2 m/s = 600 s. With Wi-Fi (40 access points, 39 hand-overs of 150 ms) each AGV is unreachable for 5.85 s per trip; at 100 commands per second that is 585 missed commands per AGV per trip, and 50 × 585 = 29,250 missed commands for the fleet – the requirement "never miss a command" is clearly violated. Even with 802.11r (about 50 ms) the loss is 39 × 50 ms = 1.95 s, i.e. 195 missed commands per AGV (9,750 for 50 AGVs). The original design cannot serve 50 moving AGVs.

C. Adaptation or Redesign – 2 Marks

Describe the revised design (for example: 5G URLLC for AGVs and robots, Wi-Fi 6 for offices, fibre backbone).

Answer

Decision: MODIFY the design – retain the wired and error-control parts, redesign the wireless part for AGVs. Private 5G URLLC for AGVs and mobile robots (4 cells of 300 m, 3 hand-overs of 20 ms, overlapping coverage); Cat6A S/FTP for fixed robots near the welders; Wi-Fi 6 with 802.11r for offices and handhelds; single-mode/multimode fibre backbone with a 10 Gbps uplink and a second link;

CCTV on PoE cabling with local clip recording; LoRa for slow sensors; strict-priority QoS for robot and AGV traffic; CRC on wired links, 5G FEC/HARQ on the radio and a Selective Repeat window of 335 frames on the head-office link.

D. Evidence-Based Defence – 2 Marks

Defend your decision with numbers and the trade-offs involved.

Answer

Numbers: time unreachable per AGV trip falls from 5.85 s to 0.06 s (−99%, 97.5 times less); missed commands per trip fall from 585 to about 6 per AGV (−99%); the uplink load of 1,473.92 Mbps (147% of 1 Gbps) falls to 14.7% on a 10 Gbps uplink (or 74% of 1 Gbps if the plant CCTV is recorded locally but the AGV cameras stream live – still too high, so 10 Gbps is chosen); 4 cells replace 40 access points along the route.

Trade-offs: private 5G costs more than Wi-Fi (spectrum licence, core network, radio units, SIM management) and needs new skills; about 6 command slots per trip can still be missed during the 20 ms hand-overs, so AGVs must have an onboard safe-stop (20 ms at 2 m/s is only 4 cm of travel) and dual connectivity or overlapping cells; Wi-Fi 6 is kept for offices to save cost. The decision to modify rather than fully redesign keeps the investment in cabling and error control while fixing the only part that fails the 50-AGV requirement.